

Showcase of Cisco Switch Automation Program

--- With Screenshots

Synopsis

A brief introduction to the program with graphical illustration. Below is the frontend:

Cisco Infrastructure Automation Console
Granular Frontend Switch Configuration, Validation, and State Lifecycle Controller

Target Connection Settings

Switch IP Address
SSH Username
SSH Password
Enable Secret

Desired Configuration Intent

Device Hostname
VLAN Profiles Database
VLAN 10
VLAN 20
VLAN 50

Operations Execution Center

Execution Stream Logs

[SYSTEM] Console logging buffer cleared.

The Target Connection Settings

Assuming we need to manipulate more than one Cisco Switch, I made the connection to be dynamic, meaning we can provide our target Switch's IP address and credential to connect a specific Cisco Switch.

I have embedded a set of default value (Switch IP address and credential), the values are changeable.

Desired Configuration Intent

Here we can determine the configurations for Device **Hostname** and **VLAN**. I have default values populated, while all these values are changeable.

Operations Execution Center

I have separated the function to four functions, and ordered them in a deliberate sequence as showing in the screenshot. You can simply strike the buttons to perform the corresponding actions.

During the implementation of each action, in the **Execution Stream Logs** section, the blank field, a real-time log stream will be showing there, which empowers us with more granular control and visualizes the execution. When an error arises, we can identify the root cause instantly.

Clear Logs

The 'Clear Logs' button only clears the logs in the frontend interface, 'Execution Stream Logs' section, providing a fresh and clean screen.

Demonstration --- Evidence of functioning/Tests

Connection handling & Deploy Configuration

Prepare the Cisco switch for the test:

```
Switch>show vlan bri
```

VLAN	Name	Status	Ports
1	default	active	Gi0/0, Gi0/1, Gi0/2, Gi0/3 Gi1/0, Gi1/1, Gi1/2, Gi1/3 Gi2/0, Gi2/1, Gi2/2, Gi2/3 Gi3/0, Gi3/1, Gi3/2, Gi3/3
1002	fddi-default	act/unsup	
1003	token-ring-default	act/unsup	
1004	fddinet-default	act/unsup	
1005	trnet-default	act/unsup	

```
Switch>int g0/0
```

Hostname: **Switch**

No VLAN configuration

```
interface GigabitEthernet0/0
no switchport
ip address 192.168.74.200 255.255.255.0
negotiation auto
!
interface GigabitEthernet0/1
media-type rj45
negotiation auto
```

Assigned 192.168.74.200/24 to Switch port g0/0 for remote access.

```
no ip http server
no ip http secure-server
!
ip ssh version 2
!
!
!
```

```
line con 0
line aux 0
line vty 0 4
  login local
  transport input ssh
!
!
```

SSH enabled.

Deploy Configuration --- successful in Frontend

Cisco Infrastructure Automation Console

Granular Frontend Switch Configuration, Validation, and State Lifecycle Controller

Target Connection Settings

Switch IP Address 192.168.74.200
SSH Username admin
SSH Password •••••
Enable Secret •••••

Desired Configuration Intent

Device Hostname SWITCH_AUTOMATED

VLAN Profiles Database

VLAN 10	VLAN_DATA
VLAN 20	VLAN_VOICE
VLAN 50	VLAN_SECURITY

⚙️ Operations Execution Center

Deploy Configuration Validate Setup Save to NVRAM Download Backup

Execution Stream Logs

Clear Logs

[SYSTEM] Console session tracking initialized. Awaiting pipeline action execution input...

```
[REQUEST] Contacting local server controller pipeline endpoint: /deploy...
[06:27:13] WARNING: * Debugger is active!
[06:27:13] INFO: * Debugger PIN: 861-102-956
[06:27:43] INFO: 127.0.0.1 - - [22/May/2026:06:27:43] "GET / HTTP/1.1" 200 -
[06:29:36] INFO: Initiating SSH deployment connection to 192.168.74.200...
[06:29:36] INFO: Connected (version 2.0, client Cisco-1.25)
[06:29:37] INFO: Auth banner: b"r\n\r\n*****\r\n\r\n IOSv is strictly limited to use for evaluation, demonstration and IOS education. IOSv is provided as a courtesy tool to help you evaluate and learn this platform. All rights reserved. No part of this program may be reproduced, stored in a retrieval system, or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, or by any information storage and retrieval system, without prior written permission from Cisco Systems, Inc."
[06:29:37] INFO: Authentication (password) successful
[06:29:37] INFO: Authentication successful. Entering configuration mode.
[06:29:37] INFO: Staging hostname change to: SWITCH01 AUTOMATED
[06:29:37] INFO: Staging VLAN 10 with name: VLAN DATA
[06:29:37] INFO: Staging VLAN 20 with name: VLAN VOICE
[06:29:37] INFO: Staging VLAN 50 with name: VLAN SECURITY
[06:29:46] INFO: Configuration commands successfully transmitted to device.
[SUCCESS] configuration deployed successfully.
```

Strike the **'Deploy Configuration'** button, clear processes are displayed in the Execution Stream Logs section. We can see Initiating SSH connection, Authentication, Configuring, and a clear message: **Configuration deployed successfully.**

From Frontend, we can see the Deployment is successful.

Verification from Kali vector, server side--- successful

```
(venv)-(kali@kali)-[~/DevOps_Proj/cisco-switch-automation]
$ python app.py
* Serving Flask app 'app'
* Debug mode: on
[06:27:13] INFO: WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on http://127.0.0.1:5000
[06:27:13] INFO: Press CTRL+C to quit
[06:27:13] INFO: * Restarting with stat
[06:27:13] WARNING: * Debugger is active!
[06:27:13] INFO: * Debugger PIN: 861-182-956
[06:27:43] INFO: 127.0.0.1 - - [22/May/2026 06:27:43] "GET / HTTP/1.1" 200 -
[06:29:36] INFO: Initiating SSH deployment connection to 192.168.74.200...
[06:29:36] INFO: Connected (version 2.0, client Cisco-1.25)
[06:29:37] INFO: Auth banner: b"\r\n*****
**\r\n* IOSv is strictly limited to use for evaluation, demonstration and IOS *
\r\n* education. IOSv is provided as-is and is not supported by Cisco's
\r\n* Technical Advisory Center. Any use or disclosure, in whole or in part,
\r\n* of the IOSv Software or Documentation to any third party for any
\r\n* purposes is expressly prohibited except as otherwise authorized by
\r\n* Cisco in writing.
*****"
[06:29:37] INFO: Authentication (password) successful!
[06:29:37] INFO: Authentication successful. Entering configuration mode.
[06:29:37] INFO: Staging hostname change to: SWITCH_AUTOMATED
[06:29:37] INFO: Staging VLAN 10 with name: VLAN_DATA
[06:29:37] INFO: Staging VLAN 20 with name: VLAN_VOICE
[06:29:37] INFO: Staging VLAN 50 with name: VLAN_SECURITY
[06:29:46] INFO: Configuration commands successfully transmitted to device.
[06:29:46] INFO: 127.0.0.1 - - [22/May/2026 06:29:46] "POST /deploy HTTP/1.1" 200 -
```

Kali is hosting the server for testing, and it shows almost the same information, indicating a successful deployment.

Verification from Switch vector--- successful

```
SWITCH_AUTOMATED>enable
Password:
SWITCH_AUTOMATED#show running-config
Building configuration...

Current configuration : 3751 bytes
!
! Last configuration change at 10:29:24 UTC Fri May 22 2026 by admin
!
version 15.2
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
service compress-config
!
hostname SWITCH_AUTOMATED
!
boot-start-marker
```

```
SWITCH_AUTOMATED#
SWITCH_AUTOMATED#
SWITCH_AUTOMATED#show vlan bri

VLAN Name                Status    Ports
-----
1    default                active    Gi0/1, Gi0/2, Gi0/3, Gi1/0
                                           Gi1/1, Gi1/2, Gi1/3, Gi2/0
                                           Gi2/1, Gi2/2, Gi2/3, Gi3/0
                                           Gi3/1, Gi3/2, Gi3/3
10   VLAN_DATA               active
20   VLAN_VOICE              active
50   VLAN_SECURITY           active
1002 fddi-default           act/unsup
1003 token-ring-default     act/unsup
1004 fddinet-default        act/unsup
1005 trnet-def               act/unsup
```

Hostname and VLANs are configured as we intended in running-config!

Some words to the Connection handling and the beauty of Stream logs

Cisco Infrastructure Automation Console

Granular Frontend Switch Configuration, Validation, and State Lifecycle Controller

Target Connection Settings

Switch IP Address

SSH Username

SSH Password

Enable Secret

Desired Configuration Intent

Device Hostname

VLAN Profiles Database

VLAN 10

VLAN 20

VLAN 50

Operations Execution Center

Execution Stream Logs

```
[SYSTEM] Console session tracking initialized. Awaiting pipeline action execution input...
[REQUEST] Contacting local server controller pipeline endpoint: /deploy...
[06:08:39] WARNING: * Debugger is active!
[06:08:39] INFO: * Debugger PIN: 861-182-956
[06:08:53] INFO: 127.0.0.1 - - [22/May/2026 06:08:53] "GET / HTTP/1.1" 200 -
[06:14:58] INFO: Initiating SSH deployment connection to 192.168.74.200...
[06:14:58] ERROR: Network deployment operation failed: BaseConnection.__init__() got an unexpected keyword argument 'ssh_config_dict'. Did you mean 'ssh_config_file'?
[API ERROR] Deployment failed: BaseConnection.__init__() got an unexpected keyword argument 'ssh_config_dict'. Did you mean 'ssh_config_file'?
```

The example above is a failure test, it fails at the connection establishment. The Execution Stream Logs displays the exact error message, which helps us to pinpoint the root cause and accelerate our debugging process.

Validate Setup

Switch IP Address 192.168.74.200

SSH Username admin

SSH Password *****

Enable Secret *****

🎯

Desired Configuration Intent

Device Hostname SWITCH_AUTOMATED

VLAN Profiles Database

VLAN 10 VLAN_DATA

VLAN 20 VLAN_VOICE

VLAN 50 VLAN_SECURITY

⚙️

Operations Execution Center

Deploy Configuration

Validate Setup

Save to NVRAM

Download Backup

Execution Stream .logs

Clear Logs

[SYSTEM] Console session tracking initialized. Awaiting pipeline action execution input...

[REQUEST] Contacting local server controller pipeline endpoint: /deploy...

[06:27:13] WARNING: * Debugger is active!

[06:27:13] INFO: * Debugger PIN: 861-182-956

[06:27:43] INFO: 127.0.0.1 - - [22/May/2026 06:27:43] "GET / HTTP/1.1" 200 -

[06:29:36] INFO: Initiating SSH deployment connection to 192.168.74.200...

[06:29:36] INFO: Connected (version 2.0, client Cisco-1.25)

[06:29:37] INFO: Auth banner: b'\r\n*****\r\n IOSv is strictly limited to use for evaluation, demonstration and IOS \r\n education. IOSv is prov:

[06:29:37] INFO: Authentication (password) successful!

[06:29:37] INFO: Authentication successful. Entering configuration mode.

[06:29:37] INFO: Staging hostname change to: SWITCH_AUTOMATED

[06:29:37] INFO: Staging VLAN 10 with name: VLAN_DATA

[06:29:37] INFO: Staging VLAN 20 with name: VLAN_VOICE

[06:29:37] INFO: Staging VLAN 50 with name: VLAN_SECURITY

[06:29:46] INFO: Configuration commands successfully transmitted to device.

[SUCCESS] Configuration deployed successfully.

[REQUEST] Contacting local server controller pipeline endpoint: /validate...

[06:29:46] INFO: 127.0.0.1 - - [22/May/2026 06:29:46] "POST /deploy HTTP/1.1" 200 -

[06:39:24] INFO: Connecting to 192.168.74.200 for operational state validation...

[06:39:24] INFO: Connected (version 2.0, client Cisco-1.25)

[06:39:24] INFO: Auth banner: b'\r\n*****\r\n IOSv is strictly limited to use for evaluation, demonstration and IOS \r\n education. IOSv is prov:

[06:39:24] INFO: Authentication (password) successful!

[06:39:25] INFO: Validating device hostname state...

[06:39:27] INFO: Validating operational VLAN database...

[06:39:27] INFO: State validation passed perfectly! Running profile fully matches intended state.

[SUCCESS] All settings validated successfully. 0 discrepancies found.

Strike **'Validate Setup'**, we can see the Stream log says 'All settings validated successfully, 0 discrepancies found. According to our verification in last section, we verified from Frontend, Kali and Switch, those information are consistent.

A further test of the **'Validate Setup'** function

--- Manually create discrepancy on Cisco Switch.

```

SWITCH_AUTOMATED(config-vlan)#do show vlan bri

VLAN  Name
-----
1    default          active    Gi0/1, Gi0/2, Gi0/3, Gi1/0
                                           Gi1/1, Gi1/2, Gi1/3, Gi2/0
                                           Gi2/1, Gi2/2, Gi2/3, Gi3/0
                                           Gi3/1, Gi3/2, Gi3/3
10   VLAN_DATA         active
20   VLAN_VOICE       active
50   VLAN_SECURITY    active
1002 fddi-default      act/unsup
1003 token-ring-default act/unsup
1004 fddinet-default  act/unsup
1005 trnet-default    act/unsup
SWITCH_AUTOMATED(config-vlan)#exit
SWITCH_AUTOMATED(config)#
SWITCH_AUTOMATED(config)#end
SWITCH_AUTOMATED#
*May 23 02:16:19.734: %SYS-5-CONFIG-I: Configured from console by console
SWITCH_AUTOMATED#vlan database
% Warning: It is recommended to configure VLAN from config mode,
as VLAN database mode is being deprecated. Please consult user
documentation for configuring VTP/VLAN in config mode.

SWITCH_AUTOMATED(vlan)#vlan 10 name Marketing
VLAN 10 modified:
Name: Marketing
SWITCH_AUTOMATED(vlan)#do show vlan bri

```

Manually updated VLAN 10's name from VLAN_DATA to **Marketing** on Cisco Switch, thereby we created one discrepancy.

Run Validate Setup again

Cisco Switch Automation
+
127.0.0.1:5000
OffSec
Kali Linux
Kali Tools
Kali Forums
Kali NetHunter
Exploit-DB
Google Hacking DB

SSH Username admin
SSH Password
Enable Secret

Desired Configuration Intent

Device Hostname SWITCH_AUTOMATED
VLAN Profiles Database
VLAN 10 VLAN_DATA
VLAN 20 VLAN_VOICE
VLAN 50 VLAN_SECURITY

Operations Execution Center

Deploy Configuration
Validate Setup
Save to NVRAM
Download Backup

Execution Stream Logs
Clear Logs

[SYSTEM] Console session tracking initialized. Awaiting pipeline action execution input...
[REQUEST] Contacting local server controller pipeline endpoint: /deploy...
[22:02:28] WARNING: * Debugger is active!
[22:02:28] INFO: * Debugger PIN: 861-182-956
[22:02:47] INFO: 127.0.0.1 - - [22/May/2026 22:02:47] "GET / HTTP/1.1" 200 -
[22:03:29] INFO: 127.0.0.1 - - [22/May/2026 22:03:29] "[3mGET /favicon.ico HTTP/1.1[0m" 404 -
[22:04:02] INFO: Initiating SSH deployment connection to 192.168.74.200...
[22:04:02] INFO: Connected (version 2.0, client Cisco-1.25)
[22:04:03] INFO: Auth banner: b"*****\n\n IOSv is strictly limited to use for evaluation, demonstration and IOS * \n\n educatio
[22:04:03] INFO: Authentication (password) successful!
[22:04:03] INFO: Authentication successful. Entering configuration mode.
[22:04:03] INFO: Staging hostname change to: SWITCH_AUTOMATED
[22:04:03] INFO: Staging VLAN 10 with name: VLAN_DATA
[22:04:03] INFO: Staging VLAN 20 with name: VLAN_VOICE
[22:04:03] INFO: Staging VLAN 50 with name: VLAN_SECURITY
[22:04:04] INFO: Configuration commands successfully transmitted to device.
[SUCCESS] Configuration deployed successfully.
[REQUEST] Contacting local server controller pipeline endpoint: /validate...
[22:04:04] INFO: 127.0.0.1 - - [22/May/2026 22:04:04] "POST /deploy HTTP/1.1" 200 -
[22:19:08] INFO: Connecting to 192.168.74.200 for operational state validation...
[22:19:08] INFO: Connected (version 2.0, client Cisco-1.25)
[22:19:08] INFO: Auth banner: b"*****\n\n IOSv is strictly limited to use for evaluation, demonstration and IOS * \n\n educatio
[22:19:08] INFO: Authentication (password) successful!
[22:19:09] INFO: Validating device hostname state...
[22:19:11] INFO: Validating operational VLAN database...
[22:19:11] WARNING: [DISCREPANCY DETECTED] Name Mismatch on VLAN 10: Expected 'VLAN_DATA', found operational name 'Marketing'.
[WARNING ALERT] Validation complete: Non-standard profile configuration patterns identified.
Mismatch detected: Name Mismatch on VLAN 10, Expected 'VLAN_DATA', found operational name 'Marketing'.

An **alert** pops up with a **warning sign**, explicitly tells 'Non-standard profile configuration patterns identified', and it explicitly points out what is the mismatch, what is the expected value and what is the detected value.

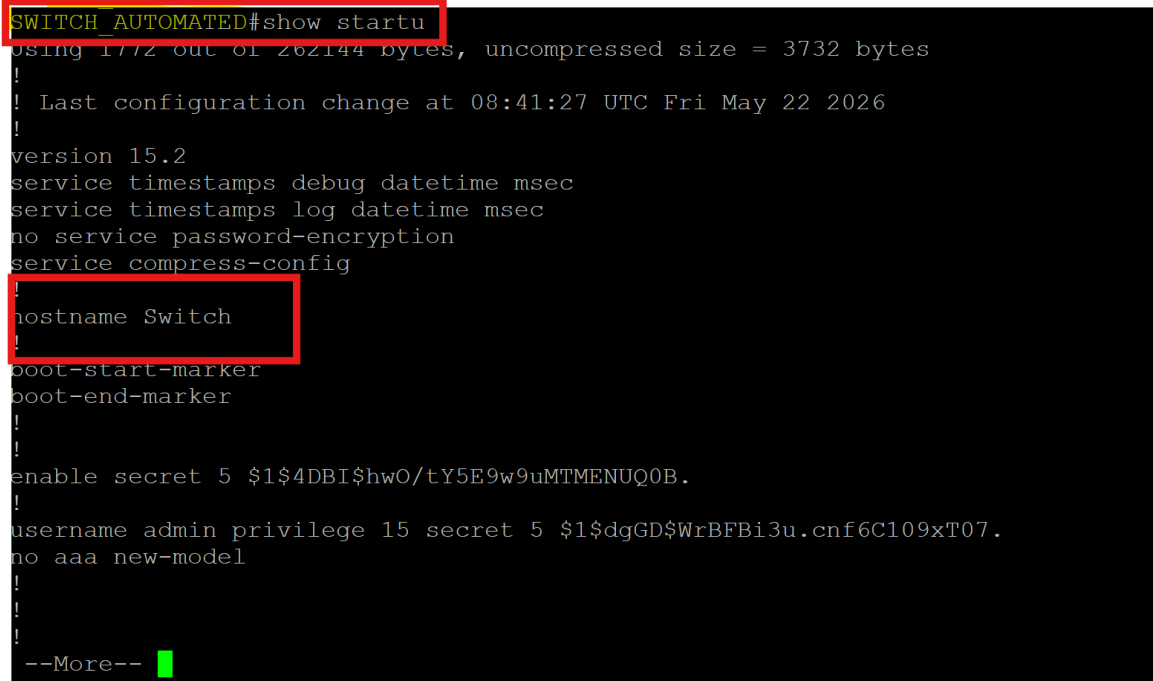
Conclusion: The 'Validate Setup' function is working as expected.

Save to NVRAM

Of startup-config, prior to Save action

issue command:

show startup-config



```
SWITCH_AUTOMATED#show startu
Using 1772 out of 262144 bytes, uncompressed size = 3732 bytes
!
! Last configuration change at 08:41:27 UTC Fri May 22 2026
!
version 15.2
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
service compress-config
!
hostname Switch
!
boot-start-marker
boot-end-marker
!
enable secret 5 $1$4DBI$hwO/tY5E9w9uMTMENUQ0B.
!
username admin privilege 15 secret 5 $1$dgGD$WrBFBi3u.cnf6C109xT07.
no aaa new-model
!
!
--More--
```

We can see the inconsistent between **running-config** and **startup-config**, meaning the configuration hasn't been saved to NVRAM yet.

'Save to NVRAM', strike the button on Frontend


```
[SYSTEM] Console session tracking initialized. Awaiting pipeline action execution input...
[REQUEST] Contacting local server controller pipeline endpoint: /deploy...
[06:27:13] WARNING: * Debugger is active!
[06:27:13] INFO: * Debugger PIN: 861-182-956
[06:27:43] INFO: 127.0.0.1 - - [22/May/2026 06:27:43] "GET / HTTP/1.1" 200 -
[06:29:36] INFO: Initiating SSH deployment connection to 192.168.74.208...
[06:29:36] INFO: Connected (version 2.0, client Cisco-1.25)
[06:29:37] INFO: Auth banner: b"\r\n*****\r\n* IOSv is strictly limited to use for evaluation, demonstration and IOS *\r\n* education. IOSv is prov
[06:29:37] INFO: Authentication (password) successful!
[06:29:37] INFO: Authentication successful. Entering configuration mode.
[06:29:37] INFO: Staging hostname change to: SWITCH_AUTOMATED
[06:29:37] INFO: Staging VLAN 10 with name: VLAN_DATA
[06:29:37] INFO: Staging VLAN 20 with name: VLAN_VOICE
[06:29:37] INFO: Staging VLAN 50 with name: VLAN_SECURITY
[06:29:46] INFO: Configuration commands successfully transmitted to device.
[SUCCESS] Configuration deployed successfully.
[REQUEST] Contacting local server controller pipeline endpoint: /validate...
[06:29:46] INFO: 127.0.0.1 - - [22/May/2026 06:29:46] "POST /deploy HTTP/1.1" 200 -
[06:39:24] INFO: Connecting to 192.168.74.208 for operational state validation...
[06:39:24] INFO: Connected (version 2.0, client Cisco-1.25)
[06:39:24] INFO: Auth banner: b"\r\n*****\r\n* IOSv is strictly limited to use for evaluation, demonstration and IOS *\r\n* education. IOSv is prov
[06:39:24] INFO: Authentication (password) successful!
[06:39:25] INFO: Validating device hostname state...
[06:39:27] INFO: Validating operational VLAN database...
[06:39:27] INFO: State validation passed perfectly! Running profile fully matches intended state.
[SUCCESS] All settings validated successfully. 0 discrepancies found.
[REQUEST] Contacting local server controller pipeline endpoint: /save...
[06:39:27] INFO: 127.0.0.1 - - [22/May/2026 06:39:27] "POST /validate HTTP/1.1" 200 -
[06:56:11] INFO: Connecting to 192.168.74.208 to save current running configuration...
[06:56:11] INFO: Connected (version 2.0, client Cisco-1.25)
[06:56:12] INFO: Auth banner: b"\r\n*****\r\n* IOSv is strictly limited to use for evaluation, demonstration and IOS *\r\n* education. IOSv is prov
[06:56:12] INFO: Authentication (password) successful!
[06:56:18] INFO: Running-configuration successfully written to NVRAM startup-config.
[SUCCESS] Running configuration saved to NVRAM storage device memory.
```

Stream logs indicating ‘Running configuration saved to NVRAM’.

```
*****"
[06:56:12] INFO: Authentication (password) successful!
[06:56:18] INFO: Running-configuration successfully written to NVRAM startup-config.
[06:56:18] INFO: 127.0.0.1 - - [22/May/2026 06:56:18] "POST /save HTTP/1.1" 200 -
```

Kali, the server side saying the same information.

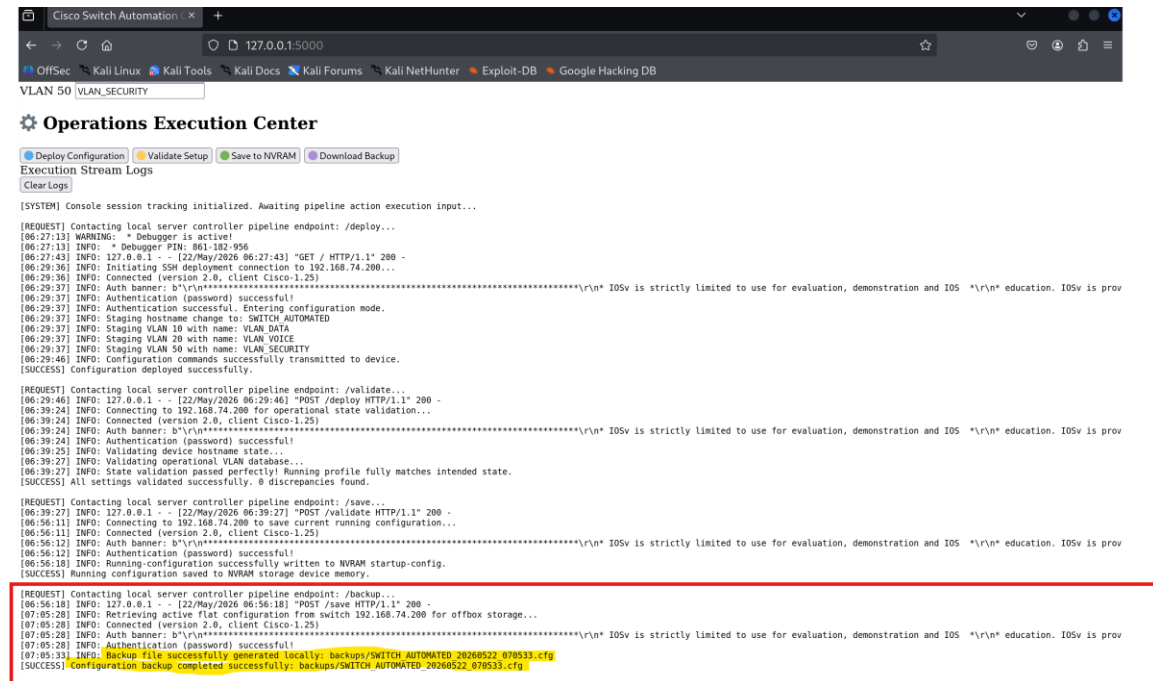
```
SWITCH_AUTOMATED#show startup-co
Using 1788 out of 262144 bytes, uncompressed size = 3751 bytes
!
! Last configuration change at 10:29:24 UTC Fri May 22 2026 by admin
!
version 15.2
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
service compress-config
!
hostname SWITCH_AUTOMATED
!
```

From switch vector, issue command ‘show startup-config’ on switch, the configuration is consistent now.

Conclusion: the ‘Save to NVRAM’ function is working as expected.

Download backup --- back up locally

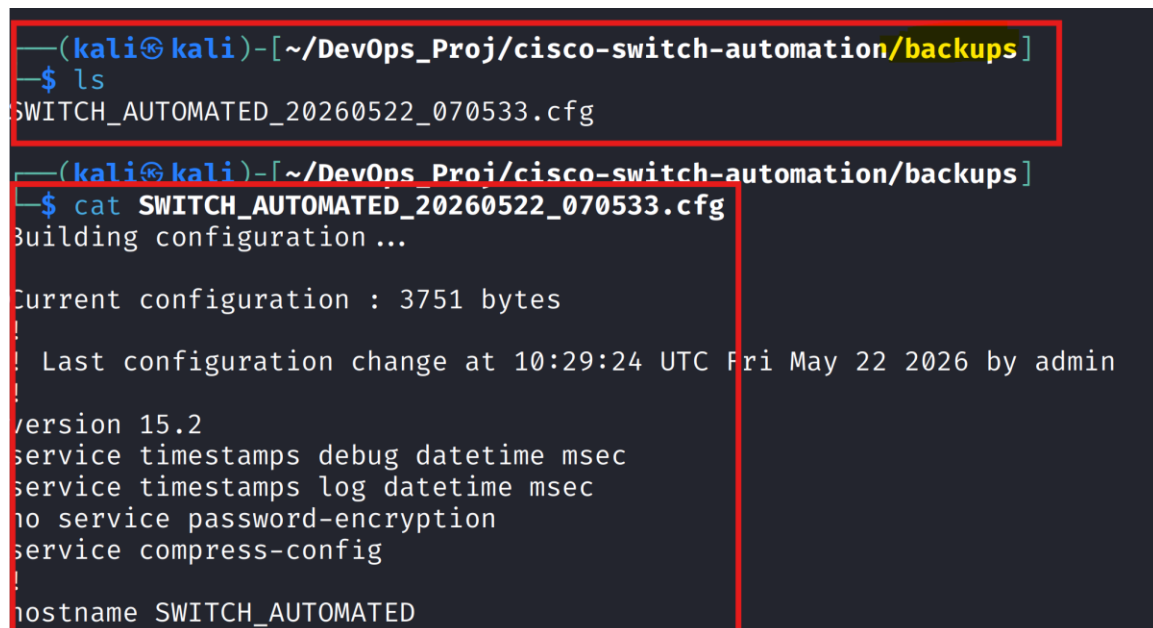
Strike 'Download backup' button



The screenshot shows the Cisco Switch Automation web interface. At the top, there's a navigation bar with links to OffSec, Kali Linux, Kali Tools, Kali Docs, Kali Forums, Kali NetHunter, Exploit-DB, and Google Hacking DB. Below this is a dropdown menu for 'VLAN 50' and a button labeled 'VLAN_SECURITY'. The main section is titled 'Operations Execution Center' and contains four buttons: 'Deploy Configuration', 'Validate Setup', 'Save to NVRAM', and 'Download Backup'. Below these buttons is a section for 'Execution Stream Logs' with a 'Clear Logs' link. The logs themselves show a sequence of events: contacting the local server controller, validating the configuration, saving it to NVRAM, and finally, backing up the configuration. The backup file is named 'backups/SWITCH_AUTOMATED_20260522_070533.cfg'.

```
[SYSTEM] Console session tracking initialized. Awaiting pipeline action execution input...
[REQUEST] Contacting local server controller pipeline endpoint: /deploy...
[06:27:13] WARNING: * Debugger is active!
[06:27:13] INFO: * Debugger PID: 861-182-956
[06:27:43] INFO: 127.0.0.1 - - [22/May/2026 06:27:43] "GET / HTTP/1.1" 200 -
[06:29:36] INFO: Initiating SSH deployment connection to 192.168.74.200...
[06:29:36] INFO: Connected (version 2.0, client Cisco-1.25)
[06:29:37] INFO: Auth banner: b'\r\n*****\r\n IOSv is strictly limited to use for evaluation, demonstration and IOS \r\n\r\n education. IOSv is prov
[06:29:37] INFO: Authentication (password) successful!
[06:29:37] INFO: Authentication successful. Entering configuration mode.
[06:29:37] INFO: Staging hostname change to: SWITCH_AUTOMATED
[06:29:37] INFO: Staging VLAN 10 with name: VLAN_DATA
[06:29:37] INFO: Staging VLAN 20 with name: VLAN_VOICE
[06:29:37] INFO: Staging VLAN 50 with name: VLAN_SECURITY
[06:29:46] INFO: Configuration commands successfully transmitted to device.
[SUCCESS] Configuration deployed successfully.
[REQUEST] Contacting local server controller pipeline endpoint: /validate...
[06:29:46] INFO: 127.0.0.1 - - [22/May/2026 06:29:46] "POST /deploy HTTP/1.1" 200 -
[06:39:24] INFO: Connecting to 192.168.74.200 for operational state validation...
[06:39:24] INFO: Connected (version 2.0, client Cisco-1.25)
[06:39:24] INFO: Auth banner: b'\r\n*****\r\n IOSv is strictly limited to use for evaluation, demonstration and IOS \r\n\r\n education. IOSv is prov
[06:39:24] INFO: Authentication (password) successful!
[06:39:25] INFO: Validating device hostname state...
[06:39:27] INFO: Validating operational VLAN database...
[06:39:27] INFO: State validation passed perfectly! Running profile fully matches intended state.
[SUCCESS] All settings validated successfully. 0 discrepancies found.
[REQUEST] Contacting local server controller pipeline endpoint: /save...
[06:39:27] INFO: 127.0.0.1 - - [22/May/2026 06:39:27] "POST /validate HTTP/1.1" 200 -
[06:56:11] INFO: Connecting to 192.168.74.200 to save current running configuration...
[06:56:11] INFO: Connected (version 2.0, client Cisco-1.25)
[06:56:12] INFO: Auth banner: b'\r\n*****\r\n IOSv is strictly limited to use for evaluation, demonstration and IOS \r\n\r\n education. IOSv is prov
[06:56:12] INFO: Authentication (password) successful!
[06:56:18] INFO: Running-configuration successfully written to NVRAM startup-config.
[SUCCESS] Running configuration saved to NVRAM storage device memory.
[REQUEST] Contacting local server controller pipeline endpoint: /backup...
[06:56:18] INFO: 127.0.0.1 - - [22/May/2026 06:56:18] "POST /save HTTP/1.1" 200 -
[07:05:28] INFO: Retrieving active flat configuration from switch 192.168.74.200 for offbox storage...
[07:05:28] INFO: Connected (version 2.0, client Cisco-1.25)
[07:05:28] INFO: Auth banner: b'\r\n*****\r\n IOSv is strictly limited to use for evaluation, demonstration and IOS \r\n\r\n education. IOSv is prov
[07:05:28] INFO: Authentication (password) successful!
[07:05:31] INFO: Backup file successfully generated locally: backups/SWITCH_AUTOMATED_20260522_070533.cfg
[SUCCESS] Configuration backup completed successfully: backups/SWITCH_AUTOMATED_20260522_070533.cfg
```

Stream logs indicates a successful backup, with the accurate file path.



The screenshot shows a terminal window on a Kali Linux machine. The user is in the directory ~/DevOps_Proj/cisco-switch-automation/backups. They run 'ls' and see the file SWITCH_AUTOMATED_20260522_070533.cfg. Then they run 'cat SWITCH_AUTOMATED_20260522_070533.cfg' and see the contents of the configuration file, which includes the version (15.2), service timestamps, and the hostname SWITCH_AUTOMATED.

```
(kali@kali)-[~/DevOps_Proj/cisco-switch-automation/backups]
$ ls
SWITCH_AUTOMATED_20260522_070533.cfg

(kali@kali)-[~/DevOps_Proj/cisco-switch-automation/backups]
$ cat SWITCH_AUTOMATED_20260522_070533.cfg
Building configuration...

Current configuration : 3751 bytes

!
Last configuration change at 10:29:24 UTC Fri May 22 2026 by admin

version 15.2
service timestamps debug datetime msec
service timestamps log datetime msec
no service password-encryption
service compress-config
!
hostname SWITCH_AUTOMATED
```

On Kali, navigate to the file path Stream logs pointed out, the backup file is present.

When 'cat' the backup file, we can see it is the configuration.

Conclusion: the configuration backup function is working as expected.